

Balan Gunasekaran

Senior GIS Analyst

ikbalan02@gmail.com

+917338961060

India, IN



Professional Summary

GIS Analyst with 6 years of experience in spatial data analysis, cartographic design, remote sensing, and geospatial database management. Expert in leveraging large scale UAV Photogrammetry datasets and management. Skilled in transforming complex datasets into actionable insights and supporting data-driven decision-making across environmental, urban planning, utilities, and infrastructure projects. Experience in ArcGIS Pro, QGIS, Python, and spatial analytics.

Work Experience

08/2025 – 02/2026 Chennai	GIS Analyst, Freelance - Assisted in wind resource assessment and data analysis using industry-standard tools and methodologies. - Worked with meteorological data (wind speed, direction, power density) for site evaluation. - Supported wind turbine performance analysis and feasibility studies. - Gained exposure to wind energy standards, policies, and testing procedures. - Participated in field measurements, data collection, and report preparation. - Learned basics of SCADA data analysis and energy yield estimation. - Collaborated with scientists/engineers on renewable energy projects. - Developed skills in technical documentation, presentations, and teamwork. - Improved understanding of renewable energy systems and sustainability practices.
06/2024 – 06/2025 Riyadh, Saudi Arabia	Data Processing Specialist_GIS, AMAI - Hands on experience in topographical and hydrological sites analyse when assessing land suitability for a solar and wind power. - Analyzing visual pollution involves identifying, mapping, and assessing features in the environment that negatively impact the aesthetic quality of an arca. - Prepared land-use maps, base maps, and zoning proposals using GIS tools (ArcGIS/QGIS/ AutoCAD) to analyze existing conditions and future development potential. - Collaborated with government authorities, architects, engineers, and community stakeholders to collect data and develop planning solutions aligned with local development policies. - Reviewed building proposals for compliance with town planning guidelines and zoning regulations. - Contributed to the preparation and revision of master Plans for urban and peri-urban areas, including land use allocation, zoning regulations, and population growth projections. - Streamlined client communication channels, reducing response time and improving retention. - Delivered post-project support, troubleshooting, and client training on GIS data usage and visualization.

01/2022 – 05/2024
Bangalore, India

GIS Project Coordinator, AEREO

- Acted as the primary POC for client communications, addressing technical queries, providing progress updates, and ensuring project deliverables met quality and timeline expectations.
- Developed Photogrammetry workflow and QC procedure for Drone data handling right from analytics, to the consolidation of final field reports for Asset Management projects as well as mining,
- Coordinated with the client on the project requirement and framing GIS & Photogrammetry workflow for the team.
- SOP Creation & Conduct the Training Modules Session.
- Developed models and tools for easier implementation of workflow in an Agisoft using python.
- Built automated pipelines for processing large geospatial datasets, including vector, raster, and LiDAR data.
- Created QA/QC automation scripts for checking attribute consistency, missing values, coordinate system conflicts, and spatial overlaps.
- Developed custom scripts for mosaic dataset creation, updating, pyramids generation, clipping, and color balancing.
- Created a custom Python script to automatically extract spot elevation points from DEM datasets for topographical and engineering analysis.
- Developed custom Python tools for batch-processing UAV datasets in Metashape, reducing manual processing time and improving workflow consistency.
- Developed interactive dashboards and visualization tools to communicate spatial risk to business stakeholders.
- Conducted comprehensive quality assurance and validation checks on data products to ensure accuracy, consistency, and defect-free delivery.
- Customized and extended GIS tools using proprietary and open-source libraries to support spatial risk modeling requirements.
- Automated data pipelines and analytics dashboards to visualize the outputs in inhouse platform.
- Managed inhouse platform for 2D and 3D visualisation using Cesium and Mapbox.
- Experience with AWS services, specifically EC2 and S3.

03/2018 – 11/2021
Chennai, India

GIS Engineer, Gyokap

- Coordinating and collecting all responses of the client requirements to enhance the workflow.
- Created the topography vector maps of the utilities as telecom, electricity and gas majorly to monitor the connectivity of the trails.
- Worked on the vectorization of the features to create a topographical map survey for the land information details.
- Client coordination and Problem solving for the technical issues to plan the better results.
- Extracting the spatial data and non spatial features to enhance the business information to the platform for the client information requirements
- Classification of hotspots such as manufacturing defects, logistic defects etc to filtered out the results to generate the reports.
- Collaborated with field teams to develop standardized data collection protocols and workflows, streamlining field operations
- Experienced in LiDAR data processing, including point cloud classification, DEM/DTM generation, and feature extraction using tools like LAStools, ArcGIS, and CloudCompare for terrain modeling and infrastructure projects.
- Responsibility for the outputs and performed the quality checks of the data in standards.

02/2021 – 09/2021
Chennai

GIS Engineer, Agira Technologies

- Extracting the Spatial data features of the business such as bars, restaurant, parks etc with the reference from satellite image in qgis for the client information requirements.
- Updated the non Spatial data details of the business like address, numbers to enhance the business information to the platform.
- Performs the quality checks, analysed and finalizes the outputs.

10/2020 – 12/2020
Chennai, India

GIS Consultant, Detect Technologies,

- Generated and estimate the Volume using UAV DTM to Compare the changes in different periods.
- Worked on the vectorization of the features to create a topographical map survey for the land information details.
- Client coordination and Problem solving for the technical issues to plan the better results.
- Conducted varoius software tests for Thermal image Processing.

02/2020 – 05/2020
Bangalore, India

Junior Coordinator, WS ATKINS,

- Coordinating and collecting all responses of the client requirements to enhance the workflow.
- Created the topography vector maps of the utilities as Telecom, Electricity and Gas majorly to monitor the connectivity of the trails.
- Performed the Quality checks, analysed and Updated the outputs in the OSM database.

Education

08/2013 – 04/2017
Tirunelveli, India

B.E Geoinformatics, Regional Campus of Anna University

- Comprehensive study of GIS principles, techniques, and in-depth knowledge of spatial data analysis, data modeling, and geospatial database management.
- Proficiency in using industry-standard GIS software, including ArcGIS and QGIS.
- Collaborated with PWD Projects.
- Participated in a GIS course survey camp during B.E studies, gaining practical experience in applying GIS principles to field data collection, analysis, and mapping,

03/2011 – 03/2013
Thiruthuraipoondi, India

HSC, St. Antony's Matric Hr.Sec School

03/2010 – 03/2011
Thiruthuraipoondi, India

SSLC, St. Antony's Matric Hr.Sec.School

skills

ARC GIS, QGIS

(Arc GIS Pro, Arc GIS online)

AWS

Photogrammetry & LiDAR Processing

(Agisoft, Pix4D, DJI Terra, Leica Mapfactory, Context Capture)

Itwin Capture

Business Development Strategies

Python

Context Capture

Oracle

Project Management Skills

Innovative Problem Solving

AutoCAD

LiDAR Classification

(Spatix, Microstation)

Google Earth Engine

Multitasking Skills

Strong Leadership Skills